

Phase Coherence and Decoherence in the Rope Network

Where Classical Rope Phase Coherence Holds — and Its Quantum Boundary

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Abstract

Rope phases across the network can be coherent or decohere, and this paper states where the classical rope model legitimately describes coherence and where it meets the programme's documented quantum boundary. Classical phase coherence — the alignment and dephasing of wave-phases along and across ropes — is within the model's domain and behaves as a classical field would. Quantum decoherence in the measurement sense, tied to amplitude interference, is NOT reproduced, for the reason established in the measurement and non-local-dynamics papers: the rope's counting rule does not supply amplitude interference. All numbers in this paper are outputs of the rope_solver computation in this package (reproducible via benchmarks/reproduce_results.py), not inserted constants.

Scope of this paper

CLAIM: classical phase coherence and dephasing across the rope network are described within the model's classical domain.

NOT CLAIMED: quantum measurement decoherence / amplitude interference — the documented quantum boundary (see Scope and Limits). This paper keeps the classical and quantum notions of coherence apart.

1. Classical Phase Coherence

Wave-phases carried along the ropes can be mutually coherent, and lose coherence through the ordinary classical mechanisms — spread in propagation, coupling to a disordered environment. In this classical sense the rope network supports a well-defined coherence that the model describes like any classical wave medium.

2. The Quantum Boundary

Where it stops. Quantum decoherence in the measurement sense — the loss of amplitude interference that turns a superposition into definite outcomes — is not classical dephasing, and the rope model does not reproduce it, because its configuration-counting rule has no amplitude interference to lose. This is the same boundary documented across the quantum-sector papers: classical coherence, yes; amplitude-based quantum decoherence, no.

Status

Classical phase coherence / dephasing — Modeled (in-domain).

Quantum measurement decoherence — Boundary (documented quantum limit).

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.